

Introduction to Mathematics and Optimization

May, 2022

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x + y^2$$

and

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 1, y \leq 1, x + 2y \geq 1\}.$$

In this problem we focus on the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C. \end{aligned} \tag{1}$$

- (a) Show that (1) is a convex optimization problem.
- (b) Explain without drawing why C is a compact set and why this implies that (1) has a solution.
- (c) Solve the optimization problem (1) using a method from the curriculum.
- (d) Write down the KKT conditions corresponding to (1).
- (e) Explain solely from the KKT conditions why $(1, 1)$ cannot be an optimal solution to (1).
- (f) Is f strictly convex?
- (g) Give an example of a convex optimization problem of the form $\min_{x \in A} h(x)$ with $h : \mathbb{R} \rightarrow \mathbb{R}$ and $A \subseteq \mathbb{R}$, which does not have a solution, but where A is bounded.

PROBLEM 2

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x^2 + xy + y^3$$

and

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 2, y \leq 2, x + y \geq 3\}.$$

In this problem we focus on the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C. \end{aligned} \tag{2}$$

- (a) Find the critical points of f .
- (b) Does f have a local minimum?
- (c) Show without a drawing that $x \geq 1$ and $y \geq 1$ if $(x, y) \in C$.
- (d) Show that f is not a convex function.
- (e) Show that f is a strictly convex function when considered as a function on C .
- (f) Show that the optimization problem (2) has a unique solution.
- (g) Solve the optimization problem (2).

PROBLEM 3

- (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x^4 + x^3 + x^2 + x + 1.$$

Show that f is strictly convex.

- (b) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$g(x) = \frac{1}{12}x^4 - \frac{5}{6}x^3 + 3x^2 + x + 1.$$

Show that g is not convex.

- (c) The previous part illustrates that g as a function on all of $\mathbb{R}$ is not convex. Give two non-empty convex open subsets $U_1, U_2 \subseteq \mathbb{R}$ with $U_1 \cap U_2 = \emptyset$, such that $-10 \in U_1$, $10 \in U_2$, and g is convex when considered as a function on U_1 and U_2 .